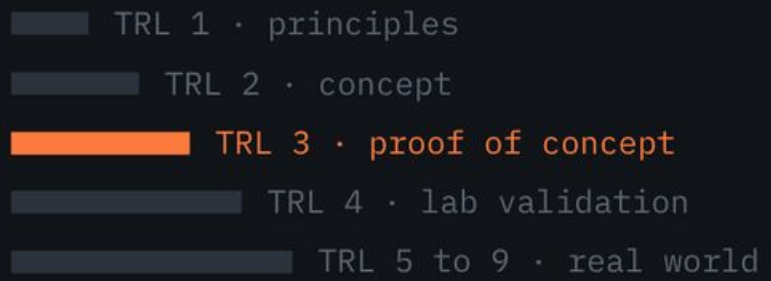
// PROOF OF CONCEPT

Prove that the technology works

A POC is like a project: It has a hypothesis, a success KPI, a deadline and a go or no-go decision.

CYRIL BAUVAIS
CO-FOUNDER AND FORMER CEO, SYNSIGHT
PHD COMPUTER AIDED DRUG DESIGN
GENOPOLE × UNIVERSITÉ D'ÉVRY

MODULE · DEEPTECH ENTREPRENEURSHIP
SESSION 2 · PART 1
12.11.2026 · 13:00



Investors need proof to invest.

how to build that proof
how to run it like a project
how to show it on one slide of your pitch

The POC comes first, before the investors.

TECHNICAL POC · TRL 3-4

Your POC happens here

BIOTECH ONLY

Clinical POC (phase 2) comes much later



Business angels and funds want to see it finished before they sign.

A finished POC is your ticket to business angels and to the seed round.

A POC answers one question: does the technology work?

TODAY · TECHNICAL PROOF

The technology works

It produces the expected result, in conditions you control, with a number you can measure.

NEXT · MARKET PROOF

Someone will pay for it

Customers, market size, willingness to pay. This is the next part of the session, with Marie Pavageau.

A good POC removes the biggest technical doubt. It does not prove the whole company.

On the NASA scale, level 3 is literally called “proof of concept”.

TRL = Technology Readiness Level, from 1 to 9



To leave level 3, NASA asks for one thing: documented experimental results that validate the predictions of the key parameters. The exit criteria are written in advance.

> **Invented at NASA in 1974. Today, the common language of R&D funding worldwide.**
>
> **NASA** 1974, 9 levels since 1995 · **US Dept. of Defense** early 2000s · **ESA** by 2008 · **EU** Horizon 2020, Horizon Europe
>
> **ISO 16290** standard, 2013 · **Bpifrance** France 2030 calls
>
> When a grant agency or an investor asks “what is your TRL?”, this is the scale they mean.

Sources: NASA ESTO, esto.nasa.gov/trl ; ISO 16290:2013 ; horizon-europe.gouv.fr, l’échelle TRL

Run your POC like any other project.

01 **An objective.** One hypothesis to prove, written in one sentence.

02 **A success KPI.** One number, measured in a known way, compared with a reference.

03 **A timeline and a budget.** A start, an end, and a check in the middle.

04 **A decision.** At the end, you go, you pivot or you stop.

Without these four elements, a POC becomes an endless research project.

Test the hardest part first.

Astro Teller runs X, Alphabet's lab for radical projects. His image: you must train a monkey to recite Shakespeare while standing on a pedestal.

THE MONKEY

Nobody knows if it is possible

This is where the project can die. Work on it first.

THE PEDESTAL

You know you can build it

It feels like progress, but it proves nothing. Keep it for later.

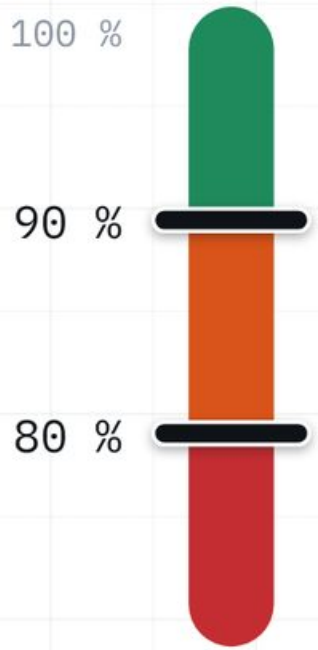
Most people start with the pedestal. In your project, what is the monkey?

Source: Astro Teller, "Tackle the monkey first", X blog, 7 December 2016



Write the decision thresholds before the first experiment.

Example: *"our model must detect the disease on images it has never seen."*



GO	Sensitivity of 90 % or more. We move to the next stage.
PIVOT	Between 80 and 90 %. We change the data, the model or the use case.
STOP	Below 80 %. We stop and we keep what we learned.

You get 78 %. You will say: “almost 80, and the data was bad that day”.

After the results, everyone finds good reasons to continue. Before, you can still be honest.

To run a POC, answer six questions.

01 / HYPOTHESIS

Critical hypothesis

What must be true for the project to exist? Test the hardest part first.

02 / EXPERIMENT

Experiment

What do we build or test, and under which conditions?

03 / KPI

Success KPI

Which number, measured how, compared with what?

04 / GO · NO-GO

Decision thresholds

Go, pivot or stop: write the thresholds before you start.

05 / RESOURCES

Time and money

How long, how much, and when is the mid-point check?

06 / STATUS

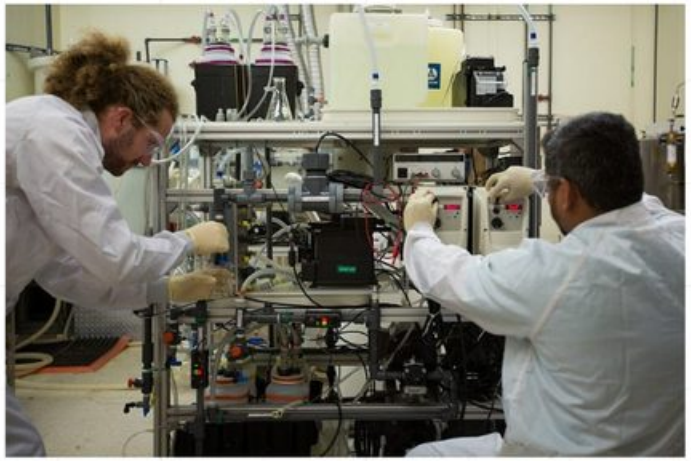
Status

What is already proven, and what remains to prove?

This is your run sheet, for you and your team (not for investors)

Foghorn: can we turn seawater into fuel as cheaply as gasoline?

Context: at Alphabet's X lab, a team tried to pull CO2 and hydrogen out of seawater and turn them into a carbon-neutral liquid fuel for cars and trucks, at a price that could compete with gasoline.



HYPOTHESIS	Extract CO2 and hydrogen from seawater and make a liquid fuel at a competitive cost.
EXPERIMENT	A full prototype: CO2 extraction, electrolysis and a methanol reactor.
KPI	Production cost per gallon of gasoline equivalent.
THRESHOLDS	8 dollars per gallon, then 5 dollars within five years.
TIME	Two years of work.

STATUS · JAN 2016

Decision: stop

The fuel was made. Cheap hydrogen was out of reach.

The technology worked, yet the project stopped: the threshold decided.

Sources: x.company/projects/foghorn · Kathy Hannun, "Three things I learned from turning seawater into fuel", X blog

Falcon 1: can a small private company put a rocket into orbit?

Context: Elon Musk founded SpaceX in 2002 to cut the cost of going to space. Falcon 1 was its first rocket, a small launcher built in-house, at a time when orbital rockets came from governments and large defence contractors.



HYPOTHESIS	A low-cost liquid-fuel rocket, built privately, can reach orbit.
EXPERIMENT	Real launches from Omelek Island, in the Pacific.
KPI	Binary and public: the rocket reaches orbit, or it does not.
FAILURES	**March 2006, March 2007, August 2008. None reached orbit.**
MONEY	Elon Musk: the fourth launch “was the last money that we had for Falcon 1”.

STATUS · 28 SEP 2008

Orbit reached

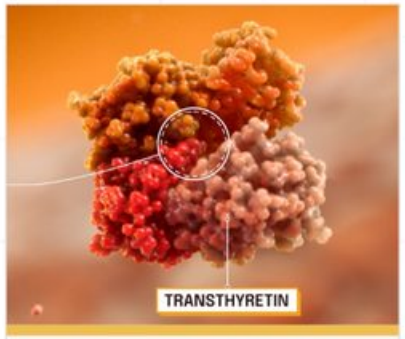
Fourth flight, 165 kg dummy payload. First commercial flight in July 2009.

A POC can fail several times. What matters is that each failure teaches you something before the money runs out.

Sources: Falcon 1 launch history, Wikipedia · Physics Today, “SpaceX rocket reaches orbit after fourth attempt”, 2008

Hereditary ATTR amyloidosis: what goes wrong.

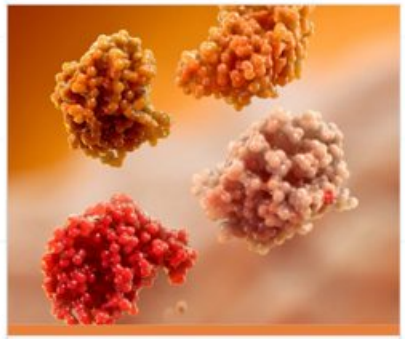
Context: hereditary transthyretin (ATTR) amyloidosis is a rare disease passed down in families. A mutation makes TTR, a protein produced by the liver, fall apart. Its pieces form amyloid deposits that slowly damage the nerves and the heart, often from adulthood.



1 · THE LIVER

Makes TTR

Four identical parts, assembled into one blood protein.



2 · A FAULTY GENE

TTR falls apart

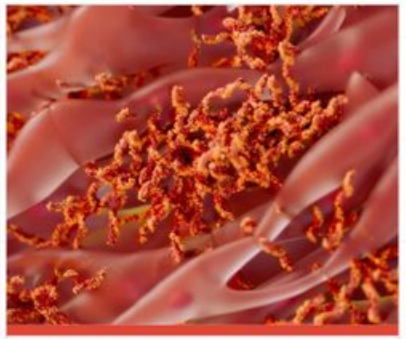
A mutation makes the four parts separate and misfold.



3 · FIBRILS

Pieces stick together

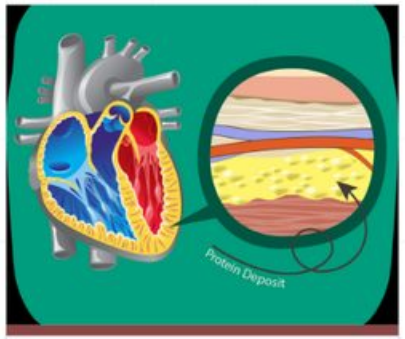
Misfolded TTR assembles into long amyloid fibrils.



4 · DEPOSITS

Fibrils invade tissue

They pile up between the cells of the nerves and the heart.



5 · THE DISEASE

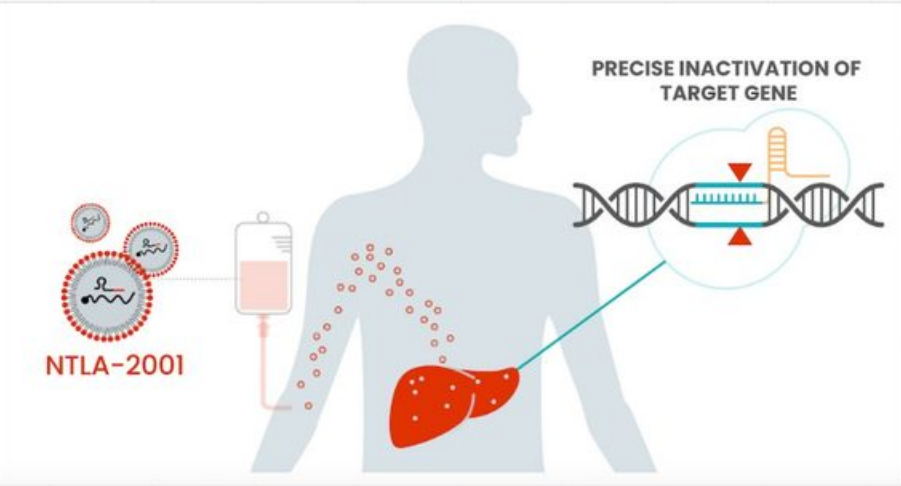
Nerves and heart fail

The heart wall stiffens. Fatal without treatment.

TODAY Drugs slow the disease but must be taken for life: daily pills or regular injections (tafamidis, patisiran, vutrisiran).

NTLA-2001: can one CRISPR injection switch off a gene inside a patient?

Context: Intellia and Regeneron tested the first CRISPR therapy injected into the bloodstream, to switch off the TTR gene directly in the liver.



HYPOTHESIS	A single infusion edits the TTR gene in the liver and lowers the TTR protein in the blood.
EXPERIMENT	6 patients with hereditary ATTR amyloidosis. Two doses.
KPI	Drop of TTR protein in the blood at day 28. Safety.
SIGNAL	A large effect that grows with the dose.
TIME	Result read 28 days after one injection.

STATUS · NEJM 2021

−52 % and −87 %

Mean TTR drop at the low and high dose. Mild side effects.

Six patients are enough when the KPI is clear. It proves the mechanism, not yet the benefit.

Source: Gillmore et al., New England Journal of Medicine 2021, doi:10.1056/NEJMoa2107454 · Diagram: Intellia Therapeutics

A model detects a disease with 99 % accuracy. Do you fund the next stage?

A team tested its model on 10,000 patients. It gave the right answer for 9,900 of them.

A

Yes

99 % is an excellent result.

B

No

This result looks suspicious.

C

I do not know yet

I need more information.



This model answers “healthy” for every patient.

Context: only 100 of the 10,000 patients have the disease. A model that always answers “healthy” is right 9,900 times.

ACCURACY

99 %

9,900 right answers out of 10,000.

SENSITIVITY

0 %

0 sick patients found out of 100.

THE MODEL, PATIENT BY PATIENT

IN REALITY	SAYS SICK	SAYS HEALTHY
100 sick patients	0	100
9,900 healthy patients	0	9,900

All 100 sick patients are missed.

A GOOD KPI FOR THIS POC

Sensitivity. Share of sick patients the model finds. Go at 90 % or more.

False alarms. Share of healthy patients flagged as sick. Go below 5 %.

Conditions. New patients, never used for training. Someone else measures.

Write the KPI and its thresholds before the first experiment.

Running a POC and pitching a POC are two different slides.

THE RUN SHEET · FOR YOU AND YOUR TEAM

Are we on track, and when do we decide?

CONTENT	Six blocks, all written before you start.
FOCUS	The decision thresholds: go, pivot or stop.
USE	Every week, to steer the work.

THE PITCH SLIDE · FOR INVESTORS

What is proven, and what will my money prove next?

CONTENT	Three zones, read in ten seconds.
FOCUS	The result so far: one number against a reference.
USE	Once, in front of people who do not know your project.

Same POC, same numbers. Only the order and the emphasis change.

Can our model read supplier invoices well enough to replace manual entry?

01 / HYPOTHESIS

Hypothesis

Our model extracts the 12 key fields of an invoice with at least 95 % accuracy.

02 / EXPERIMENT

Experiment

200 real invoices from 3 partner companies, never used for training.

03 / KPI

KPI

Share of fields extracted correctly, compared with manual entry.

04 / GO · NO-GO

Thresholds

Go at 95 % or more. Pivot between 85 and 95 %. Stop below 85 %.

05 / RESOURCES

Time and money

3 weeks, 5,000 euros. Check after 50 invoices.

06 / STATUS

Status

Done: 91 % on 20 public invoices. To do: the 200 real ones.

Step 1: the run sheet, for the team. Six blocks, real numbers, thresholds written in advance.

To pitch a POC, show three zones.

01 / PROVEN SO FAR

What is already proven?

Your best result, as one number compared with a reference. Show it big.

02 / NEXT PROOF

What will you prove next?

The experiment, the KPI and the success criterion: go at which number.

03 / WHAT WE NEED

What do you need, and what does it unlock?

Time and money for this POC, and the step it opens: a pilot, a first customer, the next round.

NO RESULT YET?

Say it plainly. Zone 2 then carries the slide.

PIVOT AND STOP THRESHOLDS?

They stay in your run sheet, ready for the questions.

The title is still your hypothesis, not the word “POC”.

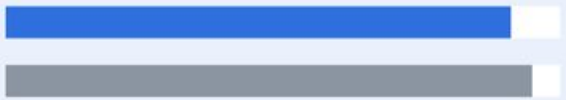
Based on investor guides: Hard2beat, 2048 Ventures, Pillar VC

Can our model read supplier invoices well enough to replace manual entry?

PROVEN SO FAR

91 %

of invoice fields extracted correctly, on 20 public invoices.



Target to replace manual entry: 95 %.

NEXT PROOF

200 real invoices

From 3 partner companies, never used for training.

KPI: share of fields extracted correctly.

GO AT 95 % OR MORE

WHAT WE NEED

3 weeks, 5,000 euros

Check after 50 invoices.

IT UNLOCKS

A paid pilot with one of the 3 partners.

Step 2: the pitch slide. Same numbers as the run sheet, with the result first.

Each group can send me two slides.

01 **The run sheet.** Six blocks: hypothesis, experiment, KPI, thresholds, time and money, status.

02 **The pitch slide.** Three zones: proven so far, next proof, what you need.

03 The title of both is your critical hypothesis, written as a question.

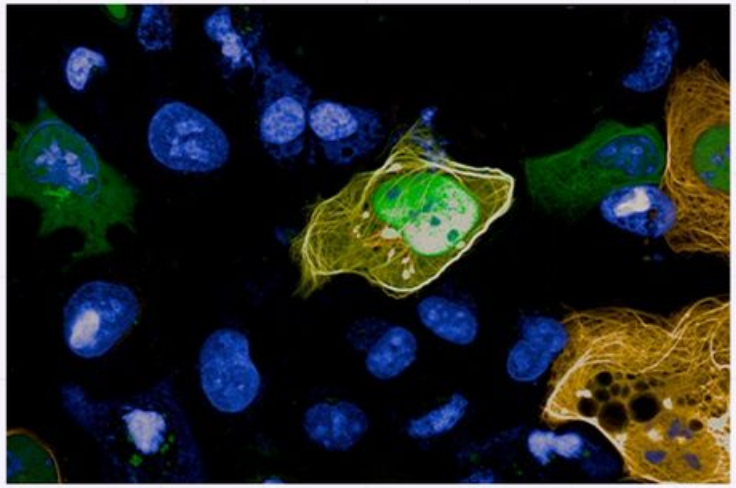
04 Real numbers. If you do not know them yet, write your best estimate.

(contact given in last slide of this deck)

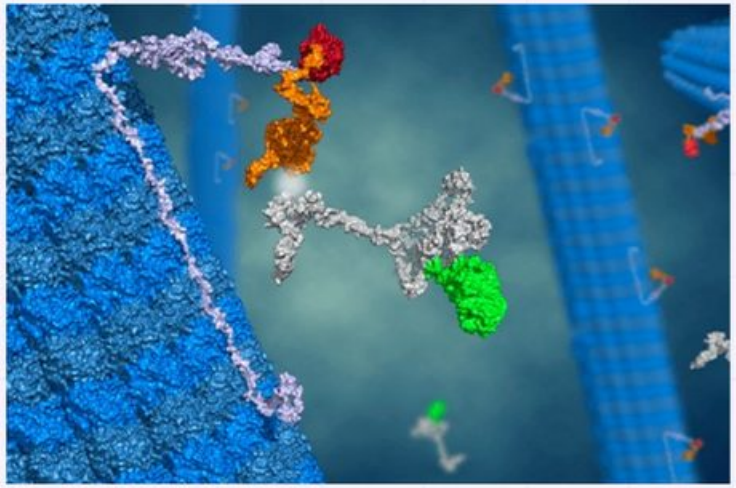
My own POC did not have the thresholds I just asked you to write.

Here is the story of Synsight, and of a POC that took much longer than planned.

Context: in 2020, Synsight took the exclusive licence on MTBench, a method invented by the SABNP lab (Inserm, Université d'Évry, Genopole) and patented by Inserm Transfert, to see interactions between proteins and RNA inside living cells.



The readout. MTBench in a cell line: interactions made visible, then measured.



The principle. Microtubules used as a bench to watch molecules bind (illustration).



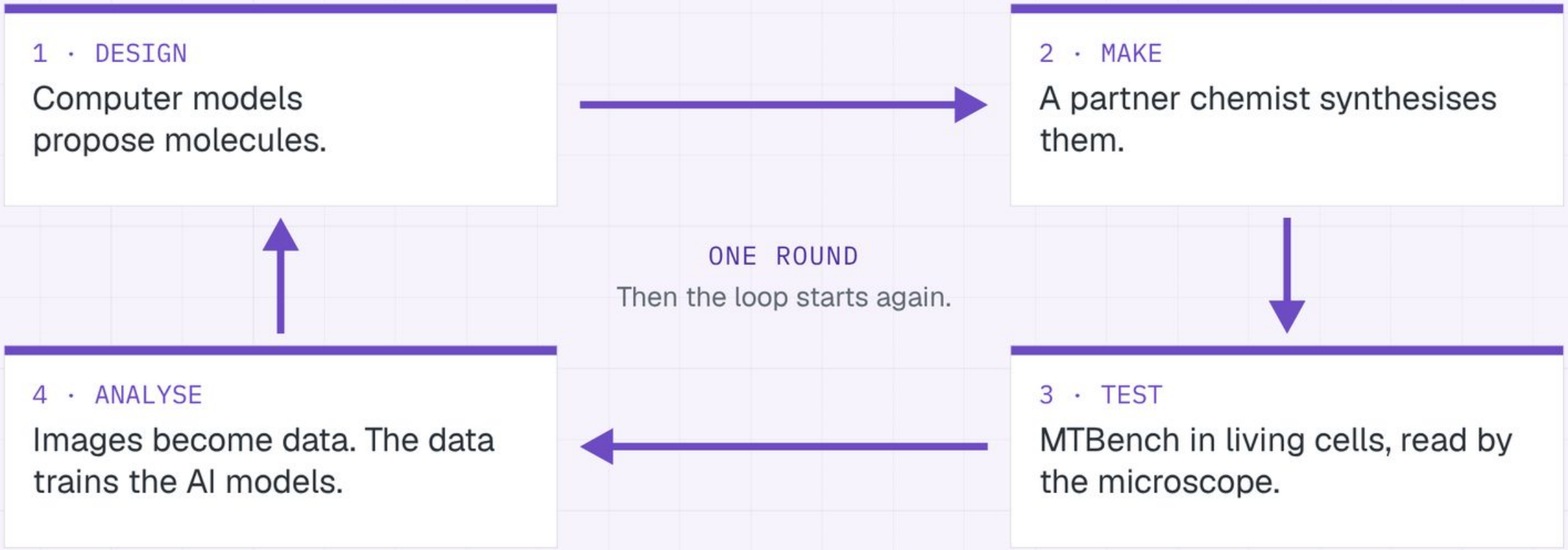
The machine. The Opera Phenix+ imager: tens of thousands of images in a few hours.

Source: genopole.fr, success story SABNP × Synsight

Synsight: AI and cell imaging to discover drugs.

2013	Two co-founders create Synsight in Évry. We design drug candidates with computer models, as a service.
THE LAB	Hosted at the University of Évry, in an Inserm lab, in the Genopole ecosystem. Access to a high-content microscope we could not afford alone.
2018-2020	We add biology and switch to a biotech model. Exclusive licence on MTBench, a patented cell technology, signed in October 2020.
2019-2022	Bpifrance, research tax credit, ANR grants and two equity rounds.
2024	Synsight is acquired by Iktos, a pioneer of generative AI for chemistry.

The microscope produced the data for the next round of molecules.



Each round makes the next molecules better. That loop is what Iktos wanted to acquire.

Can MTBench screen compounds in living cells at industrial throughput?

01 / HYPOTHESIS

Hypothesis

The assay validated at the bench keeps a clear gap between controls in 384-well plates, run by robots.

02 / EXPERIMENT

Experiment

We automated imaging, image analysis, plate feeding and pipetting. Reference system: p53/Mdm2.

03 / KPI

KPIs

Separation between positive and negative controls. (my monkey).

Compounds screened per week.

04 / GO · NO-GO

Thresholds: none

None written in advance. My bad.

05 / RESOURCES

Time

Licence signed in October 2020. Still being industrialised in 2026.

06 / STATUS · DEC 2023

1,000 compounds / week

It works, but it is not robust yet.

Enough for Synsight, which did much more. Not enough for a company built on MTBench alone.

What I learned.

01 **Write the thresholds first.** Without them, a POC never really ends. You always say “one more month”. This is called '**escalation of commitment**': the more you have spent, the harder it is to stop.

02 **Put a date on every checkpoint.** A long POC is fine if you decide at each step.

03 **“Good enough” depends on your business model.** The same result can be enough for one company and not for another. Ask what your POC is for.

Three things to remember.

01 **A POC is a project:** objective, KPI, timeline, decision.

02 **Write the thresholds before you start:** go, pivot or stop.

03 **Run slide with six blocks, pitch slide with three zones.**

A startup is many jobs, not only science.

Context: Synsight, 2013 to 2024. Two founders, both scientists. Here is what the company asked from us.

INDUSTRY
Market
Understand why pharma buys innovation.

STRATEGY
Business model
Services first, then our own technology.

LEGAL
Contracts and IP
Licence, term sheets, agreements.

FINANCE
Funding
Grants, tax credit, equity, cash plan.

INTERNATIONAL
China
Investors, partners, local rules.

PEOPLE
Team
Recruiting, advisers, outsourcing.

SALES
Partnerships
Pitch, negotiate, deliver.

M&A
Exit
Fit, negotiation, integration.

Neither of us learned this at university.

Nobody knows all of this on day one. You learn it as you go, and you get help.

Genopole is a good place to start a company.

WHAT I GOT HERE

A HOME	Offices and lab access at the University of Évry, inside an Inserm lab.
A MACHINE	A high-content microscope we could never have bought.
PEOPLE	Genopole's business advisers, and the researchers next door.









> I did not do Shaker, Genopole's incubation programme. At that time, the offer was not structured like today. But I was surrounded.

> TODAY, FOR YOU — **Shaker.** For students, PhDs and young engineers with a biotech idea: a shared lab, a mentor, business coaching.

Sources: genopole.fr, success story SABNP × Synsight · ABG, 17th edition of Shaker, June 2025

You will not do it alone. Get people around you.

The Genopole team is there for you. **On top of them**, build your own circle.

01 / ON CAMPUS

Researchers

The two inventors of MTBench joined our advisory board: an Inserm unit director and the president of the university.

02 / EXPERTS

An advisory board

A computer vision expert, a medicinal chemist with 32 years in pharma, physicians, finance advisers. We borrowed their know-how and their credibility.

03 / OTHER FOUNDERS

Réseau Entreprendre

Business owners who coach new founders, for free. They are not tech people. They know how to run a company.

15,000 volunteer business owners
2 to 3 years of coaching
Honour loans of 15 to 50 k€

You know the science. For the rest, ask people who have done it before.

Sources: Synsight decks 2022-2023 · reseau-entreprendre.org

France is a good country to start a startup.

Context: Synsight combined five sources of money. Most of it did not cost us any shares.

REVENUE

Customers

More than 800 k€ of consulting from 2013 to 2018.

BPIFRANCE

Loans and grants

Innovation loan 2019, innovation project 2020, Deep Tech aid 2022.

TAX

Research tax credit

Part of the R&D spending comes back.

RESEARCH

Research grants

An ANR project and a CIFRE PhD.

EQUITY

Investors

300 k€, then 803 k€. The founders kept 81 %.

TO START

Bpifrance is with you. Grants and loans come before any investor.

LATER

Big rounds are harder to raise. But that is a later problem.

Take the money that does not take your shares first.

Sources: Synsight company presentation 2023 · Bpifrance contracts 2019-2022

Do not be afraid to start.

01 **You are in a good place.** Genopole, the university and the labs are next door.

02 **You will find people to help you.** Advisers, researchers, other founders.

03 **France backs your first steps.** Bpifrance, grants, the research tax credit.

Synsight started in 2013 with two founders and no outside money. In 2024, it was acquired by Iktos.

Discussion.
Questions?

What is the monkey in your project?
Which result would make you stop?

Contact and resources.

- → The slides of today.
- → All the sources and the cases, plus videos to go further.
- → My contact: cyril@whitebeetle.fr

poc12112026.whitebeetle.fr